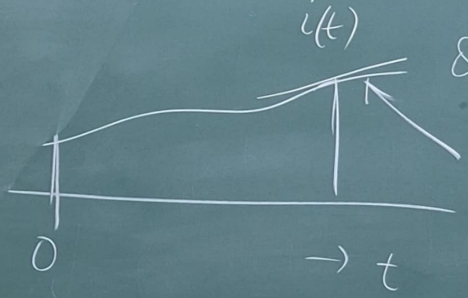
Charge

C

e 1.602×10^{-19} C

PJ1

$\int_A^B E \cdot dx$



$\frac{dQ(t)}{dt}$

$W_{AB} = \int_A^B \vec{E}(\vec{r}) d\vec{r}$

$q\vec{E}(\vec{r})$



Potential

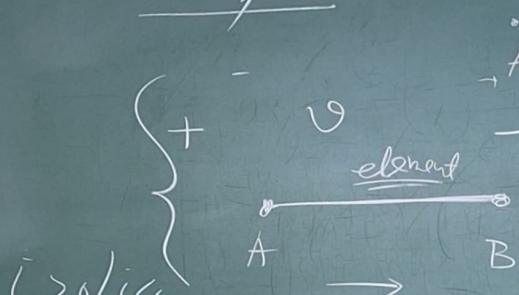
$\vec{E}(\vec{r})$

V_A : potential at A

Work to be done to bring a $1C$ charge from ∞ to A.

PSB

$v > 0 / v < 0$



$i > 0 / i < 0$

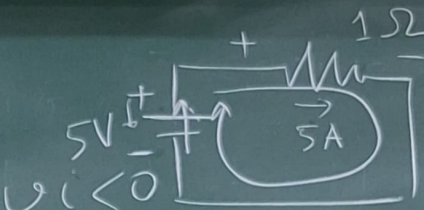
$W_{AB} = V_B - V_A$

Power

$v \cdot i > 0$ → dissipated / absorbed

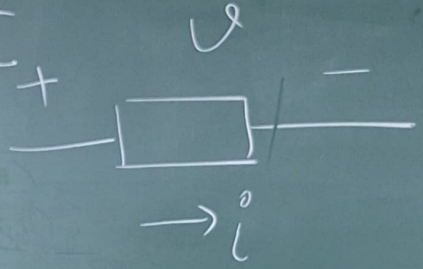
generated / released

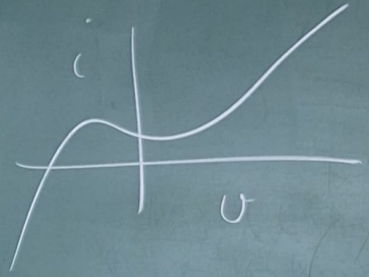
$v \cdot i < 0 \Rightarrow$ generated



25W dissipated

Lumped element

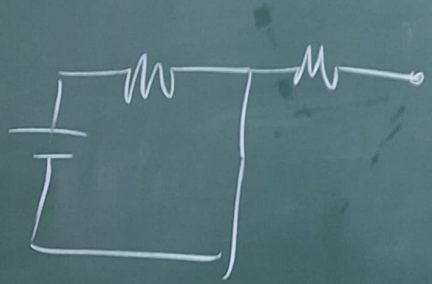




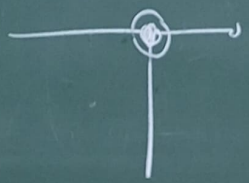
Circuit

(Graph theory formally)

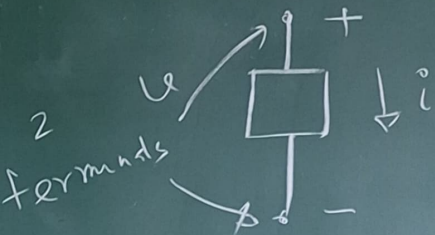
Interconnection of elements (lumped) using perfectly conducting wires

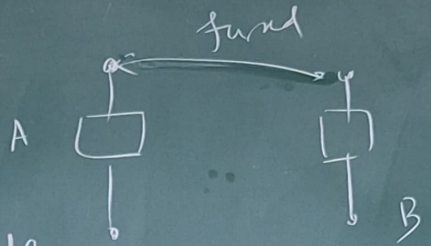


+ there is a loop! (closed)



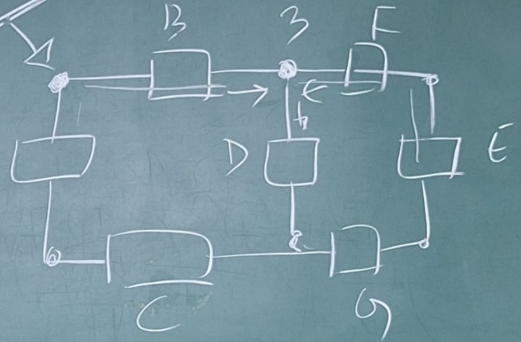
Element





(Junction) **Node**

$\geq$ 2 elements meet



(Road) **Branch** "elements"

Kirchoff's laws

Voltage law

Take any loop with elements $1, 2, \ldots 3, n, 1$

$$\oint E \, dl = 0$$ if NO magnetic field

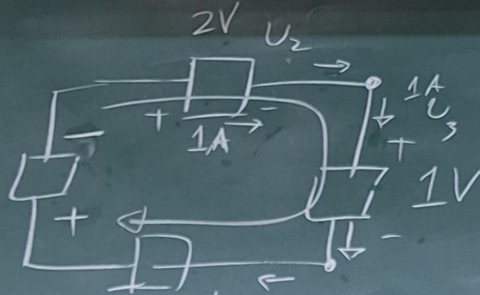
$$\sum_{i=1}^{n} v_i = 0$$

Current law

Take any node "K" & look at the elements $1, 2, \ldots m$ connected to it

Total current entering K = Total current leaving K

(No acc... of charge...



elements

$-3V = U_1$

$U = ?$

$i = ?$

$$-3 + 2 + 1 + U_4 = 0$$

$$U_4 = 0$$

Linear & Non-linear elements

KCL/KVL apply to any circuit irrespective of the nature of the elements!

This course (=) linear elements

Linear element : V–i has a linear relationship!

